Fiscal Year:	FY 2008	**Task Last Updated:**	FY 08/19/2009
PI Name:	Perchonok, Michele Ph.D.		
Project Title:	Effects of Processing and Subsequent Storage on Nutrition		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Space Human Factors Engineering		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SHFH**:Space Human Factors & Habitability (archival in 2017)		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Organization Name:	NASA Johnson Space Center		
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City:	Houston	**State:**	TX
Zip Code:	77058	**Congressional District:**	22
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	10/01/2007	**End Date:**	09/30/2008
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	NOTE: PI changed to Maya Cooper in FY2009. See Cooper for FY2009 and beyond.		
COI Name (Institution):	Cooper, Maya (NASA Johnson Space Center)		
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			
	NASA has established the goal of returning human expeditions to the moon and extending exploration to Mars. The length of proposed Mars missions increases the importance of nutritional content in the food system. Proper nutrition is essential to prevent or mitigate some physiological effects of long-term space travel as well as maintain astronaut health. The purpose of this task is to evaluate the nutritional content of the existing International Space Station (ISS) and Shuttle food system for a mission to Mars by determining the effects of thermal processing, freeze drying, irradiation, and subsequent storage time on the nutritional content of the food. In order to insure the safety of the food system and in order to deactivate metabolic enzymes that may cause the food to spoil, the packaged foods are processed to commercial sterility. This level of processing can reduce the quality of the food including nutritional content and acceptability.		

Task Description:	There has been limited measurement of the nutritional content of the flight food items. Some macronutrients and minerals are determined chemically at the Johnson Space Center Water and Food Analytical Laboratory (WAFAL). However nutrients, such as vitamins, are currently only determined through a computer program, Food Processor (ESHA Research), which calculates the combined nutritional content based on the ingredients used in the food product's formulation. The computer program provides only a close approximation of the true nutrient content and does not take into account the loss of nutrients during the thermal, freeze drying, or irradiation processes. While kinetic data have been presented in the literature for the loss of nutrition during processing, the kinetic constants provided are specific to the food and the conditions in each test. Therefore, use of the data and kinetic models in the literature will only provide a rough estimate of remaining nutritional content. Ten to twelve processed food items will be selected per year for five years and an entire nutritional profile will be determined by an off-site laboratory on each of these products at three points of its shelf life – one month after processing, at 1 year, and at 3 years - until nutritional adequacy or a need for countermeasures is established. The one year time period was chosen because it is currently the average time after production that these food items are consumed on ISS and Shuttle missions. Food items will be selected for testing based on the importance of assumed nutritional content, frequency in the menus and the type of food. Products will be stored at ambient conditions during the 3 year period to mimic the environmental storage conditions of the flight food. This study does not seek to address the effect of processing on nutrients in detail, but rather aims to place in context the overall nutritional status at the time of consumption. Data showing the effect of processing and subsequent storage will be of value pending the need to develop an "Adequate Food System" or institute countermeasures to provide health promoting foods for long duration manned space exploration.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	New project for FY2008. NOTE: PI changed to Maya Cooper in FY2009. See Cooper for FY2009 and beyond.
Bibliography Type:	Description: (Last Updated: 01/30/2012)